

Critical Review Assessment

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Paper: Li, Z., Sun, J., Murphy, L., Su, Q., Li, Z., Zhang, X., Yang, K. and Si, X. (2024) *A survey on deep learning for theorem proving*. arXiv preprint arXiv:2404.09939. Available at: <https://arxiv.org/abs/2404.09939>.

Summary

This is a survey paper that addresses the challenges in automating the formalization of mathematical proofs using deep learning, such as converting informal proofs into formal systems, generating proofs steps and selecting relevant paths, while exploring how these methods can improve theorem-proving workflows. In recent years there has been a cultural shift regarding the technical advancement of large language models that has sparked a new “wave of interest” in this area. However, there is a problem with the current research landscape that is characterized by a range of tasks, methods, datasets and evaluations that make it hard to understand true progress, to identify underlying challenges and to predict future research directions.

What this paper offers is a systematized categorization and in depth review of existing approaches spanning from auto formalization, premise selection, proof-step generation and proof research. Additionally it provides thorough summary of organized datasets and strategies for data generation, detailed analysis of evaluation metrics and performance methods, and a critical discussion of open challenges and possible future routes of exploration. “Our survey aims to serve as a foundational reference for deep learning approaches in theorem proving, inspiring and catalyzing further research endeavors in this rapidly growing field.”

The survey highlights two critical issues that current research has failed to address. Firstly, due to data scarcity, existing formal proof compilations are very limited, the Archive of Formal Proofs only contains 250 000 theorems and Lean’s math library contains 140 000. The paper says that this is “decent for small models but insufficient for models with hundreds of billions of parameters” suggesting that the field has been trying to train large language models with insufficient data. Secondly, there is the scattered evaluation, the survey shows that comparing results across different proof assistants is problematic because these automation tools work differently, for example Isabelle uses Sledgehammer (automation tool that calls external theorem provers like E and Vampire) while Lean does not have an equivalent tool. Additionally, there are other factors like resource constraints that impact results meaning a system that performs more attempts might look stronger when that might not be the case.

Ultimately, we learn from this survey that technical progress and meaningful evaluation progress are not the same. Even though we have seen huge improvements in automated theorem proving performance we can not reliably compare the results because they are achieved differently, for example with LEGO-Prover on miniF2F(57%) and Thor’s(37.3%). So in this field, where the goal is to help mathematicians, the metrics we choose could optimize the wrong thing.

With the work done on this survey we might see this area move towards more unified benchmarking practices however, the biggest change will be moving from benchmark driven to mathematician driven evaluation which will likely be a much slower transition.

There are also some open problems mentioned, one of which is how data is generated, how to create a unified evaluation system that is fair and allows for comparison, how to build tools that are optimized for mathematics instead of benchmarks and finally that AI should not only be able to prove exiting theorems but also discover new ones.

Critique

While the survey provides this systematic categorization of theorem proving approaches it does not present us with concrete solutions to solve the problems it describes. The fact that large language models are limited due to not having enough data, evaluation across proof assistants is chaotic and the misalignment between mathematician needs and benchmark driven evaluation calls for an actionable project.

With this project(Fine-tunning LLM for Mechanized Proof Automation), rather than using existing compilations of formal proofs as they come we will create a structured dataset of Isabelle proof trace pairs through a client layer. By focusing on Isabelle, we make a genuine improvement of proof automation with practical uses, however there is a trade off, while this improves data representation and tool usability, it does not address the surveys concern about a unified evaluation system across different proof assistants. The concept of extracting structured proof traces is not something new, the paper does mention datasets like LeanStep and GamePad which work in a similar way but the contribution here is the application to Isabelle and verified structured extraction.

Synthesis

Finally, this survey does a critical job, it draws a diagnosis, by systematically categorizing 180+ papers and highlighting the gaps, like the data scarcity, scattered evaluation and heavy focus on benchmarks, that allow researchers to plan actionable plans towards the landscape.

This project will respond directly to some of these issues by tackling a traceable goal of demonstrating that within a single proof assistant, with structure curated data and kernel verification we can achieve real integration that will give us practical improvements compared to what we have at the moment.